JOINT UNIVERSITIES PRELIMINARY EXAMINATIONS BOARD

AUGUST 2022 EXAMINATIONS

CHEMISTRY PRACTICAL **Time Allowed: $2^1/_2$ hours**

Answer ALL Questions.

QUESTION 2: QUALITATIVE ANALYSIS

Q is a mixture of **TWO** inorganic salts. Carry out the following tests on sample **Q**, record your test observation and inference.

(a) Transfer **(1.0g)** of **Q** into a boiling tube and add 5 ml of distilled water. Stir and allow to settle, then filter and keep the filtrate.

(b) To a small portion of the filtrate in a test tube, add small amount of ammonia solution in drops and then in excess.

(c) To the precipitate obtained in (b), add 6M HCl to the residue followed by $K_3Fe(CN)_6$

(d) To another portion of solution in (a), add HNO_3 + $AgNO_3$

(e) To the resulting mixture in (d), add NH_3 solution.

(f) To another portion of solution in (a) add HCl, then barium chloride solution.

(g) To the resulting mixture in (f), add hydrochloric acid in excess.

(h) Write an equation for the reaction taking place in (d).

(i) Provide an equation for the reaction occurring in (g).